

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026
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Course Code	24CS002PC215	Course Title	AI Assisted Coding
Year/Sem	II/I	Regulation	R24
Date and Day of Assignment	Week 7 - Wednesday	Time(s)	
Duration	2 Hours	Applicable to Batches	
Assignment Number: 13.3 (Present assignment number) / 24 (Total number of assignments)			
Q.No.	Question		Expected Time to complete
1	Lab 13 – Code Refactoring: Improving Legacy Code with AI Suggestions Lab Objectives • To introduce the concept of code refactoring and why it matters		Week 5 - Monday

	(readability, maintainability, performance). • To practice using AI tools for identifying and suggesting improvements in legacy code. • To evaluate the before vs. after versions for clarity, performance, and correctness. • To reinforce responsible AI-assisted coding practices (avoiding over-reliance, validating outputs). Learning Outcomes After completing this lab, students will be able to: 1. Use AI to analyze and refactor poorly written Python code. 2. Improve code readability, efficiency, and error handling. 3. Document AI-suggested improvements through comments and explanations. 4. Apply refactoring strategies without changing functionality. 5. Critically reflect on AI's refactoring suggestions. Task Description #1 – Remove Repetition Task: Provide AI with the following redundant code and ask it to refactor Python Code <pre>def calculate_area(shape, x, y=0): if shape == "rectangle": return x * y elif shape == "square": return x * x elif shape == "circle": return 3.14 * x * x</pre> Expected Output • Refactored version with dictionary-based dispatch or separate functions. • Cleaner and modular design. Code :	
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```
task1.py
4 def calculate_area(shape, x, y=0):
10     return area_calculators[shape]()
11
12
13 def read_file(filename):
14     with open(filename, "r") as f:
15         return f.read()
16
17
18 # Test
19 if __name__ == "__main__":
20     print(f"Rectangle: {calculate_area('rectangle', 5, 3)}")
21     print(f"Square: {calculate_area('square', 4)}")
22     print(f"Circle: {calculate_area('circle', 2)}")
23
```

•
Output:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11> & "C:/Program Files/Python313/python.exe" "c:/Users/rishi/
p/New folder/HTML/AI Ass cod/Lab 11/task1.py"
Rectangle: 15
Square: 16
Circle: 12.566370614359172
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11>
```

Task Description #2 – Error Handling in Legacy Code

Task: Legacy function without proper error handling

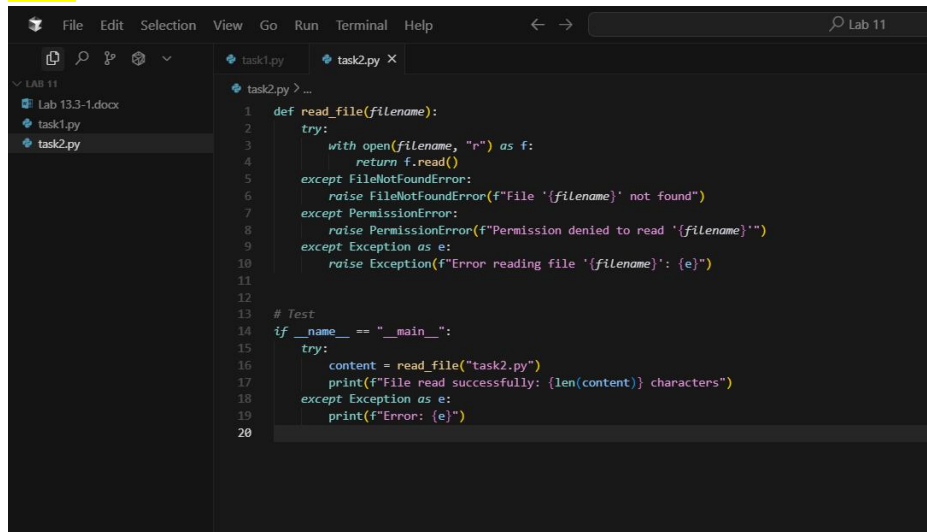
Python Code

```
def read_file(filename):
    f = open(filename, "r")
    data = f.read()
    f.close()
    return data
```

Expected Output:

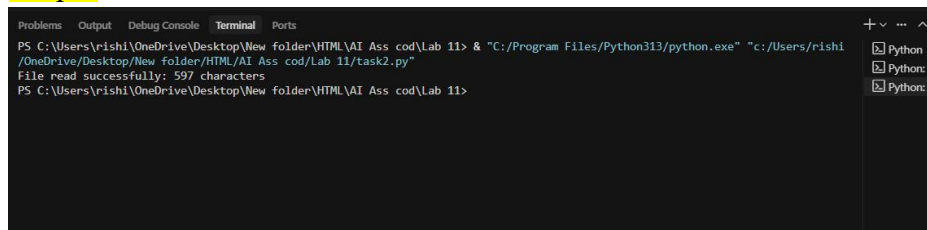
AI refactors with with open() and try-except:

Code:



```
1 def read_file(filename):
2     try:
3         with open(filename, "r") as f:
4             return f.read()
5     except FileNotFoundError:
6         raise FileNotFoundError(f"File '{filename}' not found")
7     except PermissionError:
8         raise PermissionError(f"Permission denied to read '{filename}'")
9     except Exception as e:
10        raise Exception(f"Error reading file '{filename}': {e}")
11
12
13 # Test
14 if __name__ == "__main__":
15     try:
16         content = read_file("task2.py")
17         print(f"File read successfully: {len(content)} characters")
18     except Exception as e:
19         print(f"Error: {e}")
20
```

Output:



```
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11> & "C:/Program Files/Python313/python.exe" "c:/Users/rishi/OneDrive/Desktop/New folder/HTML/AI Ass cod/Lab 11/task2.py"
File read successfully: 597 characters
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11>
```

Task Description #3 – Complex Refactoring

Task: Provide this legacy class to AI for readability and modularity improvements:

Python Code

class Student:

```
def __init__(self, n, a, m1, m2, m3):
    self.n = n
    self.a = a
    self.m1 = m1
    self.m2 = m2
    self.m3 = m3
def details(self):
    print("Name:", self.n, "Age:", self.a)
def total(self):
    return self.m1+self.m2+self.m3
```

Expected Output:

- AI improves naming (name, age, marks).
- Adds docstrings.
- Improves print readability.
- Possibly uses `sum(self.marks)` if marks stored in a list.

CODE:

```
File Edit Selection View Go Run Terminal Help
task1.py task2.py task3.py X
LAB 11
Lab 13.3-1.docx
task1.py
task2.py
task3.py
task3.py > ...
1 class Student:
2     """
3     A class to represent a student with name, age, and marks.
4     """
5
6     def __init__(self, name, age, marks):
7         """
8         Initialize a Student object.
9
10        Args:
11            name (str): The student's name
12            age (int): The student's age
13            marks (list): List of student's marks
14        """
15        self.name = name
16        self.age = age
17        self.marks = marks
18
19    def display_details(self):
20        """
21        Display student's basic information in a readable format.
22        """
23        print(f"Name: {self.name}")
24        print(f"Age: {self.age}")
25
26    def calculate_total_marks(self):
27        """
28        Calculate the total marks of the student.
29
30        Returns:
```

```
File Edit Selection View Go Run Terminal Help
task1.py task2.py task3.py X
LAB 11
Lab 13.3-1.docx
task1.py
task2.py
task3.py
task3.py > ...
1 class Student:
26     def calculate_total_marks(self):
29
30        Returns:
31            int: Sum of all marks
32        """
33        return sum(self.marks)
34
35
36 # Example usage
37 if __name__ == "__main__":
38     student = Student("John Doe", 20, [85, 92, 78])
39     student.display_details()
40     total = student.calculate_total_marks()
41     print(f"Total Marks: {total}")
42     print(f"Average: {total / len(student.marks):.2f}")
43
```

OUTPUT:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11> & "C:/Program Files/Python313/python.exe" "c:/Users/rishi
/OneDrive/Desktop/New folder/HTML/AI Ass cod/Lab 11/task2.py"
File read successfully: 597 characters
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11>
```

Task Description #4 – Inefficient Loop Refactoring

Task: Refactor this inefficient loop with AI help

Python Code

```
nums = [1,2,3,4,5,6,7,8,9,10]
squares = []
for i in nums:
    squares.append(i * i)
```

Expected Output: AI suggested a list comprehension

CODE:

```
File Edit Selection View Go Run Terminal Help
task1.py task2.py task3.py task4.py X
Lab 11
Lab 13.3-1.docx
task1.py
task2.py
task3.py
task4.py
task4.py > ...
1  nums = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
2
3  # list comprehension way - cleaner and more Pythonic
4  squares = [i * i for i in nums]
5
6  # Additional examples of list comprehensions
7  even_squares = [i * i for i in nums if i % 2 == 0]
8  cubes = [i ** 3 for i in nums]
9  strings = [str(i) for i in nums]
10
11 print(f"Numbers: {nums}")
12 print(f"Squares: {squares}")
13 print(f"Even squares: {even_squares}")
14 print(f"Cubes: {cubes}")
15 print(f"String numbers: {strings}")
16
```

OUTPUT:

```
Problems Output Debug Console Terminal Ports
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11> & "C:/Program Files/Python313/python.exe" "c:/Users/rishi
/OneDrive/Desktop/New folder/HTML/AI Ass cod/Lab 11/task4.py"
Numbers: [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]
Squares: [1, 4, 9, 16, 25, 36, 49, 64, 81, 100]
Even squares: [4, 16, 36, 64, 100]
Cubes: [1, 8, 27, 64, 125, 216, 343, 512, 729, 1000]
String numbers: ['1', '2', '3', '4', '5', '6', '7', '8', '9', '10']
PS C:\Users\rishi\OneDrive\Desktop\New folder\HTML\AI Ass cod\Lab 11>
```